

Bank of Leeds Investment Proposal

Middlesbrough SkyRail

A Suspended Monorail Network for Urban Regeneration and Economic Growth

Prepared for: Bank of Leeds – Infrastructure & Regeneration Division

Project: Middlesbrough SkyRail (Suspended Monorail)

Sector: Urban Transport, Property Development, Infrastructure, Regeneration

Investment Horizon: 25–40 years

Executive Summary

This proposal examines the business case for developing a **suspended (inverted) monorail** network serving Middlesbrough and key destinations across the Tees Valley.

Unlike a conventional metro, a suspended monorail runs beneath an elevated guideway, reducing the footprint required at street level and allowing it to pass over congested roads and junctions. Examples include the long-running suspended railway in Wuppertal, Germany, although any UK scheme would require a full engineering, planning and regulatory assessment.

The proposal should be viewed as a **long-term regeneration project**, rather than a transport project alone.

The commercial objective is to:

- stimulate city-centre regeneration,
- unlock new residential and commercial development,
- increase land values around stations,
- improve access to employment,
- attract inward investment.

Middlesbrough's current transport strategy prioritises integrated public transport, sustainable travel and supporting regeneration, but it does not currently include a monorail proposal.

Investment Thesis

Middlesbrough has:

- approximately 143,000 residents,
- a regional catchment approaching 700,000 across the Tees Valley,
- major employers,

- Teesside University,
- rail connections,
- regeneration opportunities including Middlehaven.

Despite this, much movement remains car-dependent, and improving sustainable transport is a stated objective of local transport policy.

Vision

Create Britain's first modern suspended urban monorail network.

Brand:

Middlesbrough SkyRail

Tagline:

Above the traffic. Connected to opportunity.

Proposed Network

Phase One

Approximately **18 km**

Red Line

Middlesbrough Station

↓

Town Centre

↓

Teesside University

↓

James Cook University Hospital

↓

Marton

↓

Nunthorpe

Length:

8 km

Blue Line

Middlehaven

↓

Centre Square

↓

Rail Station

↓

Riverside Stadium

↓

Tees Dock interchange

Length:

6 km

Green Line

Linthorpe

↓

Town Centre

↓

North Ormesby

↓

TeesAMP employment area

↓

Cargo Fleet

Length:

4 km

Total:

18 km

24 stations

Capital Cost Estimate

Suspended monorails vary widely in cost depending on station complexity, land acquisition, utilities and urban conditions. International urban rail projects show broad ranges, making early estimates highly uncertain.

For a conceptual estimate:

Item	Cost
Guideway	£540m
Stations	£240m
Depot	£70m
Vehicles	£120m
Power/signalling	£80m
Land & utilities	£100m
Professional fees	£90m
Contingency	£160m

Total

£1.4 billion

Rolling Stock

Fleet:

30 suspended trains

Capacity:

180 passengers

Frequency:

Every 4–5 minutes during peak periods.

Passenger Forecast

Estimated annual journeys:

Year	Passengers
1	8 million
3	10 million

Year	Passengers
5	12 million
10	15 million

Average fare:

£2.75

Annual Fare Revenue (Year 5)

12 million journeys

×

£2.75

=

£33 million

Property Development Strategy

Experience from rail and metro projects internationally suggests fixed-guideway transit can increase development interest around stations, although the scale varies significantly by market and planning policy.

Proposal:

Establish:

SkyRail Development Zones

Within:

400 metres of stations.

Potential developments:

- apartments
 - offices
 - hotels
 - retail
 - student housing
-

Commercial Property Programme

Develop:

2 million sq ft

of mixed-use property.

Investment:

£500 million

Estimated completed value:

£700 million

Retail Revenue

Station retail

24 stations

Average rent

£200,000

Annual income

£4.8 million

Advertising

Digital advertising

Naming rights

Station sponsorship

Annual revenue

£5 million

Car Parking

Park-and-ride

6 sites

Annual income

£3 million

Total Annual Operating Revenue

Source	Revenue
Fares	£33m
Retail	£4.8m
Advertising	£5m
Parking	£3m
Total	

£45.8 million

Operating Costs

Operations

£20m

Maintenance

£12m

Energy

£5m

Administration

£6m

Insurance

£2m

Total

£45 million

Observation

On these assumptions, farebox operations would be close to break-even before financing costs. That is typical of many urban transit systems, which often rely on wider economic benefits and property income rather than fare revenue alone.

Transit-Oriented Development

Bank of Leeds creates:

SkyRail Property Fund

Investment

£500 million

Build:

- apartments
- offices
- hotels

Expected long-term rental income:

£40–50 million annually.

Wider Economic Benefits

Potential outcomes include:

- higher city-centre footfall,
- improved access to employment,
- regeneration around stations,
- increased attractiveness for employers,
- enhanced university connectivity.

These wider impacts are the principal economic justification for many urban rail investments rather than direct operating profit.

Funding Structure

Source	Amount
Bank of Leeds	£400m
Institutional Investors	£400m
UK Infrastructure Funds	£400m

Source	Amount
Local/Public Funding	£200m

Total Programme

£1.4 billion

Risks

This proposal carries significant risks:

- **Demand risk:** Middlesbrough's population may not generate sufficient ridership for a high-capacity fixed-guideway system without substantial regional integration.
- **Capital cost risk:** Large urban rail projects frequently experience cost escalation.
- **Technology risk:** Suspended monorails have relatively few suppliers compared with conventional light rail.
- **Planning risk:** Elevated structures can face visual and community objections.
- **Funding risk:** Schemes of this scale would almost certainly require public-sector participation alongside private finance.

Given Middlesbrough's size, a conventional bus rapid transit or light rail system could prove more cost-effective, and any technology choice should follow a detailed options appraisal rather than assume a suspended monorail is optimal.

Strategic Recommendation

Rather than immediately pursuing a full three-line network, a more commercially realistic approach for Bank of Leeds would be:

1. Finance a detailed feasibility study and business case.
2. Develop a **single demonstration corridor** linking Middlesbrough Station, the town centre, Teesside University and James Cook University Hospital.
3. Pair the transport investment with property development around stations, allowing land-value gains and commercial real estate income to strengthen the financial case.

This phased approach reduces investment risk while providing evidence on passenger demand before committing to a larger regional network.